Between the Lines

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Introduction to Text Mining & NLP | March 2026

RESEARCH QUESTION

How has the perception of men and women in the U.S. changed over the past 25 years?

OUR CLAIM

The gap between how men's and women's lives are narrated is narrowing.

H1 Convergence

Descriptions of men and women in obituaries are converging over time

$$CosineSim(M_t, F_t) = \alpha + \beta_1 \cdot Year_t + \varepsilon_t$$

H2 Drivers

Convergence driven especially by career/profession and family/lifestyle

$$Y_i = \beta_0 + \beta_1 F_i + \Sigma \gamma_t \cdot D_t + \Sigma \delta_t \cdot (F_i \times D_t) + \varepsilon_i$$

OUR CORPUS

Text data inventory.

obituaries
~240K

time span
2001–2025

coverage
US-wide

Raw Variables

Full text, publication date, name, location

Derived / Extracted

- Gender (M/F) — first name + relational terms
- Age at death — extracted from text
- text_clean — lemmatized, names removed
- 4 thematic density scores per document
- TF-IDF vectors per year × gender

WHY OBITUARIES?

Obituaries are unguarded reflections of what a life meant.

Obituary

75, of Holliston passed away peacefully on August 21, 2023, following a brief battle with cancer. He spent his final days surrounded by family and friends.

His favorite titles were "Dad" and "Grandpa." He reveled in family dinners and eagerly seized opportunities to share his passion for science with his granddaughter.

A lifelong learner, his curiosity was evident from a young age. As a child he wanted to know how things worked; and spent his free time disassembling and rebuilding radios. This intrigue paved the way for his initial career as a technician. He went on to pursue an Engineering degree and later an MBA. He specialized in networking, and even worked on a proof of concept for the present-day internet. He spent the last two decades of his career in IT at Raytheon, a job he truly enjoyed.

Outside of work, his hobbies were a testament to his eclectic interests. He piloted Cessna planes, delved into the world of scuba diving, explored the globe with his daughter, indulged in movies, and tinkered at his electronics work bench. Amid these pursuits, he treasured time spent with lifelong friends.

He was a man of many talents and will be deeply missed.

He is survived by his daughter, Alaine, son-in-law Brennan, and his beloved granddaughters Page and Skye.

He is also survived by his extended family and friends.

A private burial and Celebration of Life will be held at a later date.

Funeral arrangements are under the care of the Ginley Crowley Funeral Home, 3 Barber Street in Medway. (ginleyfuneralhomes.com).

01 Unfiltered

Writer is not performing for a researcher — bypasses social desirability bias

02 Structured Genre

Consistent template: family, career, hobbies, faith — comparable across time

03 Dual Authorship

Captures what society, via the writer, thinks is important at the time of writing

Grimm 2010; Fowler 2007; Hume 2000

METHODOLOGY

A sequential analytical pipeline.

1 Scraping

~240K obituaries
currentobituary.com

→

2 Preprocessing

Clean, classify gender, lemmatize, remove names

→

3 EDA

Distributions, genre structure validation

Steps 1–3 form the foundation for both hypotheses

4a H1 — Overall Convergence

TF-IDF → centroids → cosine similarity → OLS

$$CosineSim = \alpha + \beta_1 \cdot Year + \varepsilon$$

→

4b H2 — Domain Drivers

Dictionary density + Female × Year interactions

$$Y = \beta_0 + \beta_1 F + \Sigma \gamma_t \cdot D_t + \Sigma \delta_t \cdot (F \times D_t) + \varepsilon$$

PREPROCESSING

How we turn raw text into structured data.

Sample Obituary

[redacted], 75
of Holliston passed away peacefully on August 21, 2023.

His favorite titles were "Dad" and "Grandpa." He reveled in family dinners and seized opportunities to share his passion for science with his granddaughter.

A lifelong learner… He went on to pursue an Engineering degree and later an MBA. He spent the last two decades of his career in IT at Raytheon.

He piloted Cessna planes, delved into scuba diving, explored the globe with his daughter, and tinkered at his electronics work bench.

He is survived by his daughter, [redacted], son-in-law [redacted], and his beloved granddaughters [redacted] and [redacted].

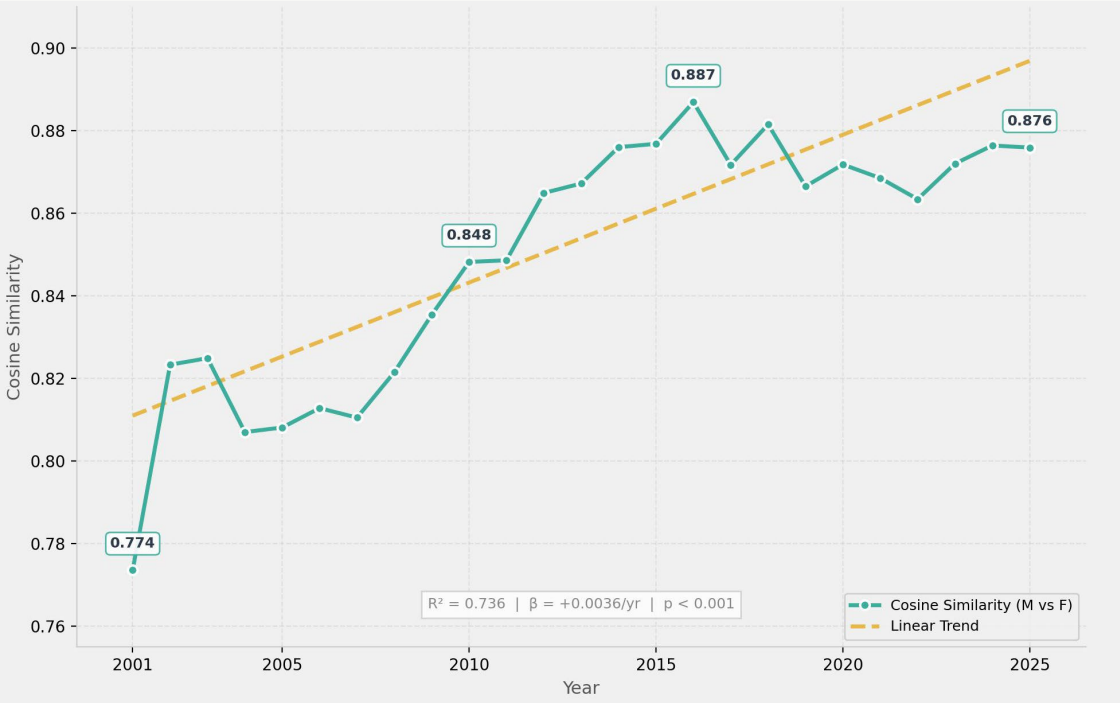
● Family ● Professional ● Lifestyle

Pipeline Steps

- ✕ Remove names, dates, places
- ✕ Strip memorial/funeral details
- ◆ Classify gender (name + relational terms)
- ◆ Lemmatize text, remove stop words
- ● Detect thematic terms via dictionaries
- ● Compute TF-IDF vectors per year × gender

FINDING 1

The language gap is closing.



How we got here

After preprocessing, for each year:

1. Fit year-specific TF-IDF vectorizer
2. Create male and female centroid vectors (mean DTM)
3. Compute cosine similarity between centroids
4. Regress similarity scores on year (OLS)

$R^2 = 0.736$

74% explained

$p < 0.001$

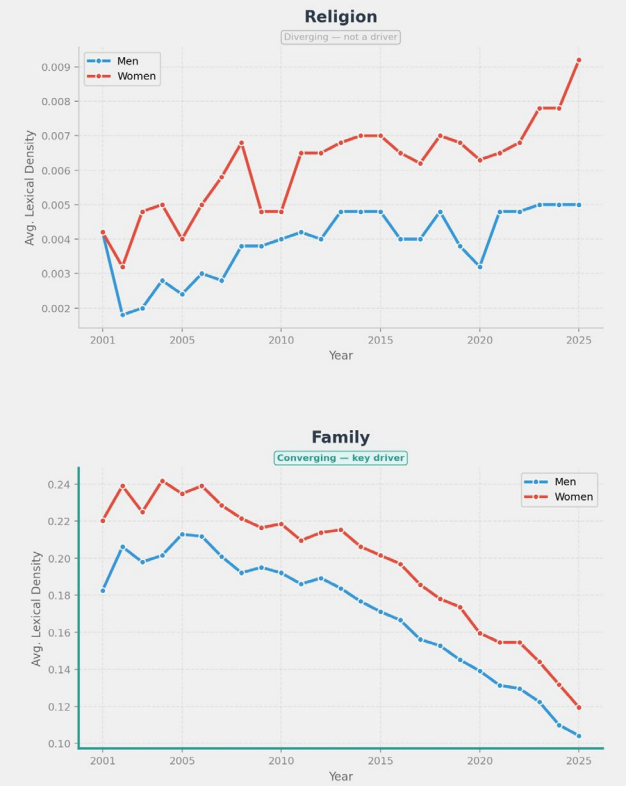
Significant

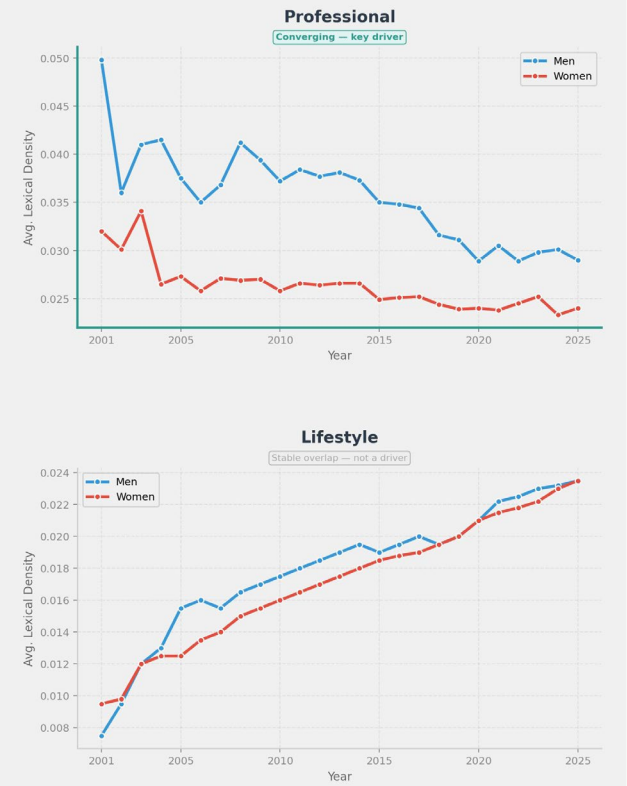
25 years | $F = 64.11$ | $\beta_1 = +0.0036/yr$

Cosine similarity rose from 0.774 in 2001 to 0.876 in 2025 — a steady, significant convergence.

FINDING 2

Not all themes converge equally.





Dictionary-based lexical density, min age 70

NEXT STEPS

Where we go from here.

1

Refine Dictionary Lexicons

Validate theme-based word lists using LDA topic outputs from the full corpus to ensure comprehensive coverage.

2

Further Refine H2 Regression

Run regressions with obituary-year fixed effects.

3

Finalize Paper & Visualizations

Polish charts for the final report, strengthen integration between motivation, theory, and findings.

4

Between the Lines

Thank you. Questions?

APPENDIX A1

Data & Scraping Sources

SO WHAT: Corpus credibility: ~240K obituaries from a US-wide aggregator ensures geographic and temporal breadth.

Source: currentobituary.com
(US-wide obituary aggregator)

Web scraper — no API available.

Each record contains:
- Full obituary text
- Publication date
- Name of deceased
- Location (city, state)

Sample raw record structure


```
{
  "name": "Robert E. Behler",
  "date": "2023-08-21",
  "location": "Holliston, MA",
  "text": "Robert E. Behler, 75,
   of Holliston passed away
   peacefully on August 21..."
}
```

APPENDIX A2

Preprocessing Technical Details

SO WHAT: Removing names, dates, and memorial details prevents genre noise from dominating the signal.

Pipeline steps:

1. HTML/special character cleaning
2. Name removal (prevents gender leakage)
3. Age extraction from text
4. Gender classification:
   name-based + relational term validation
5. Memorial detail removal:
   funeral logistics inflated religion scores
6. Lemmatization + stop word removal

Before → After example

BEFORE:
"Robert E. Behler, 75, of Holliston passed away peacefully on Aug 21, 2023. Funeral services will be held at St. Mary's Church on Friday."

AFTER:
"pass away peacefully work engineer degree MBA career raytheon pilot cessna dive explore globe daughter survive"

APPENDIX A3

Assumptions + COVID Diagnostics

SO WHAT: Min-age-70 filter ensures comparable life arcs. COVID spike visible but does not break the trend.

Key assumption:
Death is approximately random
across gender within each year.

COVID 2020–2021:
- Male mortality disproportionately higher
- Visible as dip in similarity scores
- Does not break the overall 25-year trend

Filter: min age 70 at death
→ ensures full life arcs are compared

COVID dip in convergence plot

Year	Similarity	
2018	0.882	
2019	0.867	← pre-COVID
2020	0.872	← COVID
2021	0.869	← COVID
2022	0.863	← recovery
2023	0.872	← bounce back

Dip is small (~1.5%) and temporary.
Trend resumes post-2022.

APPENDIX A4

Category Model & Dictionary Documentation

SO WHAT: Concrete, observable terms avoid subjectivity. Validated against Empath and expert review.

Religion (33 terms):
church, faith, god, jesus, christ,
lord, pastor, priest, bible, prayer...

Professional (42 terms):
career, retired, worked, professor,
engineer, manager, business, degree...

Family (31 terms):
family, mother, father, daughter,
husband, wife, survived, children...

Lifestyle (40 terms):
travel, hiking, golf, gardening,
music, cooking, photography...

Validation approach

3-step validation:

1. Empath cross-check:
   Pre-trained ~200 categories
   mapped to our 4 pillars

2. Expert review:
   Manual scan of top TF-IDF terms
   per gender per decade

3. Concrete terms only:
   Observable words, not
   sentiment or inference

APPENDIX A5

"Why Not LDA?" Comparison

SO WHAT: Short documents + genre-dominated vocab make LDA unreliable. Dictionaries give interpretable, stable results.

LDA limitations for obituaries:

1. Short docs (150–300 words)
   → unreliable topic distributions

2. Incoherent topics:
   LDA mixed military, family, memorial into single topics

3. Genre vocabulary dominance:
   "brother", "sister", "wife" appeared in nearly every topic

4. Low coherence scores across
   k = 5, 10, 15, 20 topic models

Dictionary advantages

Why dictionaries work better:

✓ Interpretable by design
(we know exactly what each category measures)

✓ Stable across years
(same vocabulary, comparable density scores)

✓ Validated externally
(Empath + expert review)

✓ Genre-aware
(memorial terms excluded)

APPENDIX A6

Full Regression Specs & Coefficients

SO WHAT: H1 confirmed: year alone explains 74% of convergence. H2: Professional gap shrank from -0.018 to -0.005.

H1: $CosineSim = \alpha + \beta_1 \cdot Year + \varepsilon$

$R^2 = 0.736$ | Adj $R^2 = 0.724$
$F = 64.11$ | $p < 0.001$
const = -6.353 (se=0.900)
$\beta_1 = +0.0036$ (se=0.000)

H2 Professional:
β_1(Female) = -0.0178 (p=0.001)
δ_{2025}(F×2025) = +0.0128 (p=0.025)
Remaining gap: ~-0.005

H2 Family regression

H2 Family:
β_1(Female) = +0.0378 (p=0.032)

Interaction trend:
$\delta_{2013} = -0.0062$
$\delta_{2018} = -0.0157$
$\delta_{2025} = -0.0225$

Remaining gap in 2025:
0.0378 - 0.0225 = 0.0153

Family gap halved but persists
(consistent with genre norms)

APPENDIX A7

Early EDA & Discarded Approaches

SO WHAT: Obituary length and age distributions are stable across gender, confirming comparability.

EDA confirmed:

- Obituary length: comparable across genders (median ~200 words)
- Age-at-death: similar distributions after min-70 filter
- Gender ratio: roughly 50/50 across all years
- Geographic spread: all 50 states represented

Discarded approaches

What we tried and dropped:

✕ N-gram analysis:
Too noisy, no clear signal

✕ Sentiment analysis:
Too coarse for this genre
(obituaries are uniformly positive)

✕ Word embeddings (Word2Vec):
Corpus too small per year
for reliable year-specific models

✕ Raw word counts:
Document length confounded results

APPENDIX A8

Stanford Empath Robustness Check

SO WHAT: Empath's pre-trained categories confirm our dictionary trends, ruling out coding-choice artifacts.

Empath: pre-trained lexicon model
~200 categories via word embeddings
(Fast et al., 2016)

Mapping to our 4 pillars:
- Religion → "religion", "worship"
- Professional → "work", "business"
- Family → "family", "wedding"
- Lifestyle → "leisure", "sports"

Result: directional trends match
our custom dictionaries

Empath vs Dictionary comparison

Theme	Dict	Empath
Professional	✓ ↓gap	✓ ↓gap
Lifestyle	✓ ↑both	✓ ↑both
Family	✓ ↓slow	✓ ↓slow
Religion	✓ flat	✓ flat

All 4 directions confirmed.
Magnitudes differ slightly
(Empath categories are broader)
but convergence pattern holds.